

Communication-Aware Adaptive Weights for Consensus-Oriented Distributed KLA-Based LMB Fusion

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ABSTRACT

Distributed multisensor, multi-object tracking over peer-to-peer networks requires conservative fusion because cross-correlations are unknown. Fixed consensus weights, however, cannot reflect changing posterior quality or the measurements actually delivered. This paper introduces a communication-aware weighting rule for the geometric-average (GA) realization of Kullback–Leibler average (KLA) fusion of labeled multi-Bernoulli (LMB) posteriors. Each node evaluates available neighbors using posterior concentration, realized link quality computed from delivery outcomes, and Bernoulli existence confidence. Separate spatial and existence branches then use these cues according to their fusion roles. Covariance and link quality govern Gaussian spatial fusion, whereas existence confidence governs Bernoulli support. An optional Fisher information distance and Fisher information accumulation (FID-FIA) cue acts only on the existence branch. The resulting Balanced mode uses the low-overhead branch backbone. Cardinality-critical mode activates the FID-FIA refinement when target-number agreement is the main risk. In an eight-sensor scenario with finite fields of view and heterogeneous packet loss, both modes reduce network disagreement relative to Fixed Metropolis and the internal adaptive baselines. They also outperform two explicitly labeled GA-LMB adaptations of related methods. Truth-referenced local accuracy is retained as a safeguard. These results frame adaptive GA-LMB fusion as a tradeoff among spatial agreement, cardinality agreement, and computation.

1. Introduction

Distributed multisensor, multi-object tracking is central to surveillance, autonomous systems, and cooperative perception. We study peer-to-peer tracking with partial field-of-view (FOV) coverage and changing posterior quality. Communication is constrained by bandwidth limits, packet drops, and temporary outages [1]. Random finite sets (RFSs), finite-set statistics (FISST), and labeled-RFS filters represent target number, kinematic state, and identity within a Bayesian framework [2–6]. The remaining question is how to combine neighboring labeled posteriors when sensing geometry, posterior uncertainty, and packet delivery vary across the network.

Within this family, generalized labeled multi-Bernoulli (GLMB) and labeled multi-Bernoulli (LMB) filters balance modeling fidelity against computational cost [5–9]. For distributed fusion, the geometric-average (GA) realization of Kullback–Leibler averaging (KLA) provides a conservative logarithmic opinion pool. This rule, also known as generalized covariance intersection (GCI), is useful when unknown cross-correlations preclude exact Bayesian fusion [10–15]. GA/KLA-based LMB fusion is therefore a natural peer-to-peer backbone. The unresolved design question is how to allocate its weights online.

Fixed Metropolis, uniform, or degree-based consensus weights are easy to deploy [16]. They cannot distinguish precise from ambiguous posteriors, nor can they represent the delivery quality realized on each link. A single adaptive score is also too coarse for an LMB posterior. A Bernoulli component may be spatially sharp but uncertain in existence, or decisive in existence but spatially diffuse. A factorized rule is therefore needed to allocate separate branch weights under communication constraints.

This paper addresses that gap with communication-aware weight allocation for distributed GA-LMB fusion. Figure 1 summarizes the update at one peer. The design combines covariance quality, realized link quality, and existence confidence in a shared backbone. It then separates the spatial and existence branches of the LMB fusion rule. A weak

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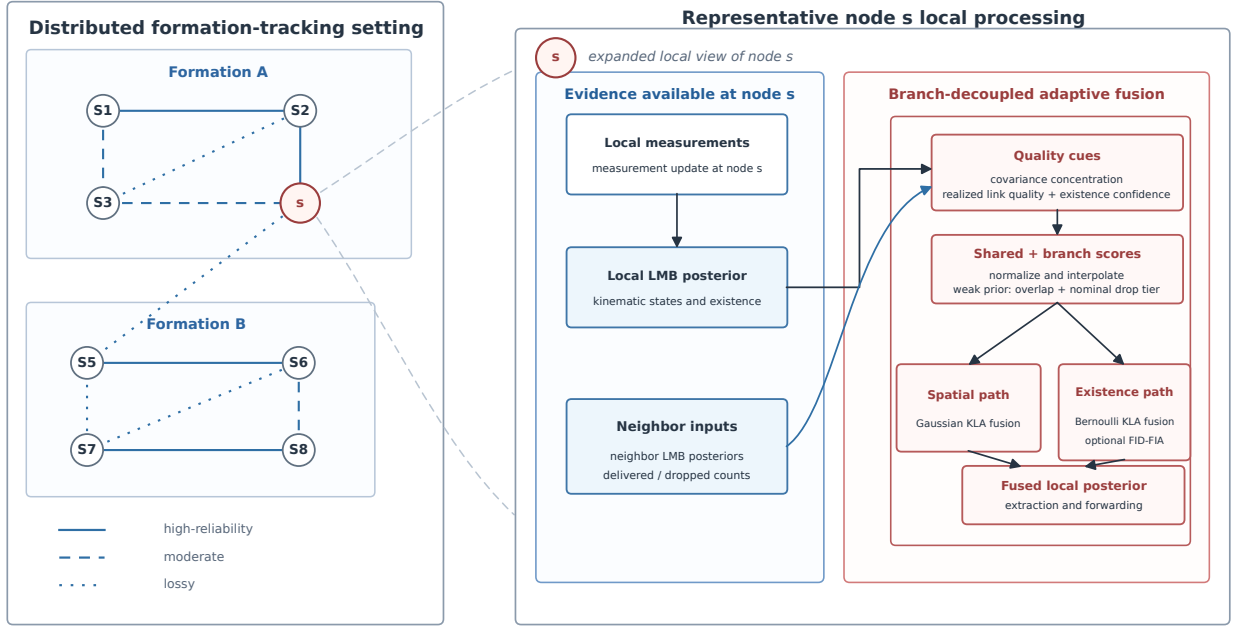


Figure 1: System overview of the distributed eight-sensor tracking setting and the update at a representative node. The current-step covariance, realized-delivery, and existence cues form a shared active-neighbor score; branch-specific interpolation is followed by a weaker prior encoding neighborhood overlap and nominal drop-tier reliability. The resulting weights enter separate Gaussian spatial and Bernoulli existence paths, and the optional FID-FIA cue acts only on the latter.

graph prior regularizes spatial fusion while remaining secondary to posterior quality. An FID-FIA refinement, adapted from Cao and Zhao [17], acts only on Bernoulli existence. This design yields Balanced mode for localization-sensitive use and Cardinality-critical mode for target-number agreement.

We evaluate the method in an eight-sensor formation scenario with variation induced by geometry and FOV, together with heterogeneous packet delivery. Network disagreement is the primary outcome because the method targets agreement in the fused multisensor picture. Truth-referenced local metrics and filter runtime guard against hidden losses in accuracy or deployability. We also compare against explicitly labeled GA-LMB adaptations of two related methods whose original filter interfaces or threat models differ from ours. The results expose the intended tradeoff. The three-factor backbone provides most of the low-cost gain, whereas FID-FIA is most useful when cardinality agreement is the dominant risk.

The main contributions of this paper are as follows.

1. A communication-aware reliability backbone for adaptive GA/KLA-LMB fusion weights under heterogeneous packet delivery.
2. A branch-decoupled LMB fusion-weight rule that separates spatial and existence weighting and confines FID-FIA to the Bernoulli existence branch.
3. A consensus-centered evaluation under packet loss and ideal links, including four communication levels, runtime measurements, GOSPA, and two literature-method adaptations.

2. Related Work

The most relevant prior work falls into three groups: labeled-RFS filtering, conservative distributed density fusion, and adaptive or weighted RFS/LMB fusion. The first group provides the statistical state representation. Mahler's RFS/FISST formulation and probability hypothesis density (PHD) recursion establish the set-valued Bayesian foundation [2–4]. GLMB and LMB filters add labeled multi-object posteriors with tractable representations of track identity [5–9]. This paper fuses LMB posteriors.

The second group concerns conservative fusion when cross-correlations are unknown. Covariance intersection, Kullback–Leibler averaging, geometric-average fusion, and inverse-covariance-intersection variants avoid overconfident fusion under unknown dependence [10–13, 18–20]. Logarithmic opinion pools and exponential mixtures connect this principle to distributed Bayesian fusion and consensus [14, 15]. GCI/KLA fusion has also been extended to multi-Bernoulli and labeled-RFS densities, including label mismatch and peer-to-peer resilience [21–23]. Gao et al. use each receiving node’s local LMB density as a trusted reference [23]. Their method tests label-associated Bernoulli components from neighbors and reduces confidence as discrepancy from the local reference increases. This attack-resilient objective differs from communication-quality weighting, but its local-trust mechanism provides a relevant comparison.

The third group is most directly related to adaptive weighting. Centralized multiple-view LMB fusion shows that information-aware weights can help when sensors have limited or different fields of view [24, 25]. Klupacs et al. fuse label-associated Bernoulli densities and tune each sensor’s contribution by object detectability [26]. Distributed LMB tracking has also used automatic track-level weight evolution across different fields of view [27]. Zheng et al. move closer to our setting by factorizing each local RFS density into subdensities [28]. Local coefficients are assigned before GA or arithmetic-average (AA) fusion under consensus or flooding. Cao and Zhao provide the closest information-geometric GA/KLA precedent [17]. Their heterogeneous labeled-RFS method uses Fisher information distance and Fisher information accumulation (FID-FIA) to assign scalar fusion weights. AA density fusion and related minimum-information-loss formulations emphasize different objectives from GA fusion [29–32]. Recent multirate work also uses Fisher-information-proportional weights after time alignment in both AA and GA fusion [33].

Communication constraints provide the deployment motivation for the weighting problem. Bandwidth limits, delays, asynchronous updates, and packet losses are central design factors in distributed fusion over sensor networks [1]. The present work focuses on the packet-delivery side of this problem by feeding realized communication quality directly into the fusion weights.

Against this background, this paper focuses specifically on online weight allocation within peer-to-peer GA/KLA-based LMB fusion. Relative to fixed-weight GA/KLA fusion, the weights react to posterior concentration and realized packet delivery. Relative to adaptive scalar-weight formulations, the rule separates the spatial and existence branches of the LMB posterior and confines the optional FID-FIA cue to the existence path.

3. Problem Formulation

Consider a distributed network with sensor set $\mathcal{V} = \{1, \dots, S\}$ and discrete times $k \in \{1, \dots, T\}$. The multi-object state at time k is the labeled random finite set X_k . Each sensor $s \in \mathcal{V}$ acquires a local measurement set $Z_k^{(s)}$. Under the standard RFS/FISST formulation, each node maintains an LMB approximation to its local posterior [2, 4–7]. Its Bernoulli components jointly encode target existence, kinematic state, and identity. The fusion layer must combine neighboring posteriors without assuming that their cross-correlations are known.

3.1. Local LMB posterior model

Let $\mathcal{N}_s \subseteq \mathcal{V}$ denote the communication neighborhood of sensor s , including the node itself. Sensor geometry and communication range induce the graph used to construct $\mathcal{N}_s$. Each node solves a local fusion problem over only these indices. The next subsection defines the delivered measurement stream and statistics for every neighbor. Prediction, measurement update, and association on stream $j \in \mathcal{N}_s$ produce one LMB posterior for fusion. We denote its finite parameter set by $\Theta_k^{(j)}$,

$$\Theta_k^{(j)} = \left\{ \left(r_{k,i}^{(j)}, p_{k,i}^{(j)}(\cdot, \ell_i) \right) \right\}_{i=1}^{M_k}, \quad (1)$$

where $\mathcal{L}_k = \{\ell_1, \dots, \ell_{M_k}\}$ is the aligned label set and M_k is its size. The quantity $r_{k,i}^{(j)} \in [0, 1]$ is the Bernoulli existence probability. The function $p_{k,i}^{(j)}$ is the spatial density for label ℓ_i . The corresponding LMB density is the standard labeled-RFS density induced by this parameter set,

$$\pi_k^{(j)}(X) = \Delta(X) w_k^{(j)}(\mathcal{L}(X)) \prod_{(x, \ell_i) \in X} p_{k,i}^{(j)}(x, \ell_i), \quad (2)$$

with

$$w_k^{(j)}(L) = \prod_{\ell_i \in L} r_{k,i}^{(j)} \prod_{\ell_i \in \mathcal{L}_k \setminus L} (1 - r_{k,i}^{(j)}), \quad L \subseteq \mathcal{L}_k. \quad (3)$$

Here $\Delta(X)$ is the distinct-label indicator and $\mathcal{L}(X)$ is the set of labels present in X . The compact parameter-set notation $\Theta_k^{(j)}$ is used below whenever the implementation operates on the Bernoulli components rather than on the full finite-set density. Thus the fusion layer below consumes LMB posterior parameters; raw measurement sets enter only through the local measurement update and the realized delivery statistics.

3.2. Communication-constrained delivery

The communication model used in the experiments acts on measurement streams before distributed LMB fusion. At time k , the raw measurement set $Z_k^{(j)}$ generated by sensor j is converted into a delivered set $\tilde{Z}_k^{(j)}$ by a communication layer with three possible effects:

1. a global or per-step bandwidth budget that limits how many measurements can be delivered,
2. a stochastic packet-drop model on each sensor stream,
3. optional node-level outages.

For node s , the local fusion problem is assembled from the delivered streams $\{\tilde{Z}_k^{(j)} : j \in \mathcal{N}_s\}$ and the matching delivery statistics. Prediction and measurement update on these streams produce the posteriors $\{\pi_k^{(j)} : j \in \mathcal{N}_s\}$ that are fused in the GA-LMB step. The delivery statistics also define the availability and link-quality factors used by the adaptive weights. Let $m_k^{(j)} \in \{0, 1\}$ denote the availability indicator of neighbor index j at time k , and let $\rho_k^{(j)} \in [0, 1]$ denote the realized delivery-quality statistic derived from delivered-versus-dropped measurements.

3.3. Distributed KLA fusion objective

Because cross-correlations among neighboring LMB posteriors are unknown, exact Bayesian fusion is generally unavailable. We therefore use the geometric-average realization of Kullback–Leibler averaging as the main conservative fusion rule. This rule is also known as generalized covariance intersection. At node s , the target density fused over $\mathcal{N}_s$ is

$$\pi_k^{(s)}(X) \propto \prod_{j \in \mathcal{N}_s} \left(\pi_k^{(j)}(X) \right)^{\omega_{k,s}^{(j)}}, \quad \sum_{j \in \mathcal{N}_s} \omega_{k,s}^{(j)} = 1, \quad \omega_{k,s}^{(j)} \geq 0, \quad (4)$$

where $\omega_{k,s}^{(j)}$ is the weight assigned by node s to neighbor j [12, 13]. Some fusion literature uses KL-averaging terminology more broadly. Here, KLA/GA denotes the geometric-average logarithmic pool in (4). Appendix-only arithmetic-average (AA) diagnostics follow the reverse-KL, minimum-information-loss route [29–31]. Fixed fusion commonly uses uniform or Metropolis consensus weights [16]. Our problem is to adapt these weights to posterior quality and realized communication quality.

In the GA-LMB fusion rule considered here, the spatial and existence parts of each Bernoulli component are fused separately after moment matching. This separation makes it natural to ask whether the same scalar weight should govern both spatial fusion and existence fusion.

3.4. Adaptive weight-allocation target

The adaptive layer returns weight vectors on the active-neighbor simplex. Fixed fusion uses one vector $\omega_{k,s}^{(j)}$ for the entire posterior. The proposed rule instead returns a spatial vector $\omega_{k,s}^{x,(j)}$ and an existence vector $\omega_{k,s}^{r,(j)}$. Both vectors are normalized over available neighbors. They may respond differently because Gaussian spatial fusion and Bernoulli existence fusion use distinct algebraic updates. Section 4 defines the scores and algorithm; this section only specifies the required outputs.

3.5. Evaluation target

The aim is to improve both local tracking and agreement among nodes. This distinction is central to the problem. Under unknown cross-correlations and communication losses, distributed fusion should move neighboring nodes toward a shared multisensor picture.

Accordingly, the evaluation uses two complementary metric layers:

- primary network-level disagreement metrics, which quantify consensus error among the fused outputs of different sensors,
- secondary truth-referenced local tracking metrics, which verify that consensus-error reductions preserve conventional tracking accuracy.

Let $\hat{X}_k^{(s)}$ denote the extracted multi-object estimate at sensor s and time k , and let $\hat{n}_k^{(s)} = |\hat{X}_k^{(s)}|$ denote its estimated cardinality. The paper defines the following network-level disagreement metrics:

$$D_{\text{OSPA}}(k) = \frac{2}{S(S-1)} \sum_{1 \leq a < b \leq S} d_{\text{OSPA}}(\hat{X}_k^{(a)}, \hat{X}_k^{(b)}), \quad (5)$$

$$D_{\text{GOSPA}}(k) = \frac{2}{S(S-1)} \sum_{1 \leq a < b \leq S} d_{\text{GOSPA}}^{(c,p,\alpha)}(\hat{X}_k^{(a)}, \hat{X}_k^{(b)}), \quad (6)$$

$$D_{\text{loc}}(k) = \frac{2}{S(S-1)} \sum_{1 \leq a < b \leq S} d_{\text{loc}}(\hat{X}_k^{(a)}, \hat{X}_k^{(b)}), \quad (7)$$

$$D_{\text{card}}(k) = \frac{1}{S} \sum_{s=1}^S \left| \hat{n}_k^{(s)} - \text{median}(\hat{n}_k^{(1)}, \dots, \hat{n}_k^{(S)}) \right|, \quad (8)$$

Here, d_{OSPA} is the optimal subpattern assignment (OSPA) distance between two estimated finite sets [34]. The quantity d_{GOSPA} is its generalized, unnormalized counterpart [35, 36]. The quantity d_{loc} is localization disagreement after Hungarian matching of common extracted objects [37]. We report these quantities as OSPA consensus error, GOSPA consensus error, matched localization disagreement, and cardinality dispersion.

All Euclidean OSPA and GOSPA quantities use cutoff $c = 5$ and order $p = 2$; GOSPA additionally uses the standard $\alpha = 2$. Both metrics use the complete extracted kinematic state vector as their Euclidean ground space, rather than position alone. For sets $X = \{x_1, \dots, x_n\}$ and $Y = \{y_1, \dots, y_m\}$ with $n \leq m$, the implementation evaluates

$$d_{\text{GOSPA}}^{(c,p,\alpha)}(X, Y) = \left[\min_{\pi} \sum_{i=1}^n \min(c, \|x_i - y_{\pi(i)}\|_2)^p + \frac{c^p}{\alpha}(m-n) \right]^{1/p}, \quad (9)$$

with the arguments exchanged when $n > m$. Unlike OSPA, this expression is not divided by $\max(n, m)$. Additional extracted objects therefore cannot reduce the score through cardinality normalization. Both scores are averaged over unordered sensor pairs, then over time and paired trials. Euclidean OSPA (E-OSPA) compares one node's extracted set with ground truth. It uses the same $(c, p) = (5, 2)$ parameters and differs from consensus OSPA only in the reference set.

These network measures use standard ingredients. OSPA and GOSPA are established finite-set metrics, and Hungarian matching provides the assignment primitive [34–36]. Pairwise dispersion is a standard measure of disagreement in consensus problems [16]. Our specific choice is the aggregation target. The metrics compare post-fusion outputs across nodes in addition to comparing each node with ground truth. This target matches the central question: whether heterogeneous communication prevents nodes from reaching a common fused picture.

The secondary local metrics remain the standard truth-referenced tracking quantities:

- local Euclidean OSPA (E-OSPA),
- local root mean square error (RMSE),
- local cardinality error.

The evaluation is asymmetric by design. Success primarily means lower network disagreement. Local metrics verify that this gain does not cause a clear loss in conventional tracking quality.

4. Communication-Aware Adaptive KLA Fusion

This section describes the adaptive fusion rule used by the distributed GA-LMB filter. The starting point is conservative geometric-average Kullback–Leibler averaging for neighboring posteriors with unknown cross-correlations

[12, 13]. The method allocates online fusion weights from bounded reliability cues. Each node then passes these weights to its GA-LMB merger. Appendix C gives variational and information-geometric interpretations of the separated branches.

4.1. Baseline GA-LMB fusion

At node s , let $\mathcal{N}_s$ denote the local communication neighborhood. Let $\pi_k^{(j)}$ be the measurement-updated LMB posterior density formed from the delivered stream associated with neighbor $j \in \mathcal{N}_s$, as described in Section 3. The conservative fusion target is the weighted geometric average in (4). The remaining task is to specify how the LMB parameter sets are moment-projected and how the spatial and existence weights are consumed by the merger.

After moment projection, the Gaussian approximation of the i th Bernoulli spatial density at node j is

$$p_{k,i}^{(j)}(x) \approx \mathcal{N}\left(x; v_{k,i}^{(j)}, T_{k,i}^{(j)}\right). \quad (10)$$

Using spatial fusion weights $\omega_{k,s}^{x,(j)}$, the fused spatial density is obtained in canonical form:

$$K_{k,i}^{(s)} = \sum_{j \in \mathcal{N}_s} \omega_{k,s}^{x,(j)} \left(T_{k,i}^{(j)}\right)^{-1}, \quad h_{k,i}^{(s)} = \sum_{j \in \mathcal{N}_s} \omega_{k,s}^{x,(j)} \left(T_{k,i}^{(j)}\right)^{-1} v_{k,i}^{(j)}, \quad (11)$$

$$\Sigma_{k,i}^{(s)} = \left(K_{k,i}^{(s)}\right)^{-1}, \quad \mu_{k,i}^{(s)} = \Sigma_{k,i}^{(s)} h_{k,i}^{(s)}. \quad (12)$$

Using existence fusion weights $\omega_{k,s}^{r,(j)}$, the Bernoulli existence probability is fused by the corresponding weighted geometric rule:

$$r_{k,i}^{(s)} = \frac{\eta_{k,i}^{(s)} \prod_{j \in \mathcal{N}_s} \left(r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}}}{\eta_{k,i}^{(s)} \prod_{j \in \mathcal{N}_s} \left(r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}} + \prod_{j \in \mathcal{N}_s} \left(1 - r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}}}, \quad (13)$$

where $\eta_{k,i}^{(s)}$ is the Gaussian normalization term induced by the fused spatial density.

4.2. Adaptive reliability factors

The adaptive layer is organized as a reliability-allocation design. It first forms a shared active-neighbor backbone that identifies informative fusion inputs. The next subsection routes this backbone into branch-specific spatial and existence weights.

Shared backbone. For node s and neighbor $j \in \mathcal{N}_s$, define the raw score

$$\tilde{\omega}_{k,s}^{(j)} = m_k^{(j)} \cdot q_{\text{cov},k}^{(j)} \cdot q_{\text{link},k}^{(j)} \cdot q_{\text{exist},k}^{(j)}, \quad (14)$$

where $m_k^{(j)} \in \{0, 1\}$ is the availability mask, $q_{\text{cov},k}^{(j)}$ is a covariance-quality score, $q_{\text{link},k}^{(j)}$ is a realized link-quality score, and $q_{\text{exist},k}^{(j)}$ is an existence-confidence score. The normalized weight is

$$\bar{\omega}_{k,s}^{(j)} = \frac{\tilde{\omega}_{k,s}^{(j)}}{\sum_{u \in \mathcal{N}_s} \tilde{\omega}_{k,s}^{(u)}}. \quad (15)$$

This factorization addresses the three failure modes most directly tied to distributed fusion: diffuse local posteriors, missing delivered information, and ambiguous Bernoulli support. Figure 2 shows how this shared backbone feeds the decoupled spatial and existence-weight paths used by the two operating modes.

Remark 1 (backbone rationale). The product form is a compact way to combine independent reliability cues on the simplex. In log form, it adds the utilities $\log q_{\text{cov}}$, $\log q_{\text{link}}$, and $\log q_{\text{exist}}$, so a poor value in any cue cannot be completely hidden by a large value in another cue. The availability mask $m_k^{(j)}$ is a hard gate for an unavailable neighbor, whereas the positive floors used in some cue mappings are soft safeguards for uncertain but still usable posteriors. If all active raw scores vanish, the implementation falls back to the uniform distribution over active neighbors before applying any smoothing or final floor.

Posterior concentration and realized delivery. For node j , let $T_{k,i}^{(j)}$ be the moment-matched covariance of the i th Bernoulli component. The covariance-quality score is defined as

$$q_{\text{cov},k}^{(j)} = \frac{1}{\epsilon + \frac{1}{M_k} \sum_{i=1}^{M_k} \text{tr}(T_{k,i}^{(j)})}, \quad (16)$$

where $\epsilon > 0$ is a small numerical constant used to avoid division by zero. A node with smaller posterior spread therefore receives a larger score.

The link-quality term is derived from realized communication outcomes. Let $d_k^{(j)}$ and $\ell_k^{(j)}$ denote the counts of delivered and dropped measurements, respectively. The realized link-quality score is defined as

$$q_{\text{link},k}^{(j)} = \frac{d_k^{(j)}}{d_k^{(j)} + \ell_k^{(j)}}, \quad (17)$$

which directly reflects how much of that node's measurement stream remains available after delivery constraints.

Design rationale. The covariance cue is a local posterior-concentration proxy: a smaller average trace means that, after moment projection, the neighbor contributes a sharper Gaussian component to the canonical-form spatial average above. The link cue uses delivered and dropped measurements at the current time step, so the weights reflect the information available to the local fusion problem.

Existence decisiveness. The third cue is separated because Bernoulli existence uncertainty differs from spatial posterior spread. For the i th Bernoulli component at node j ,

$$c_{k,i}^{(j)} = |2r_{k,i}^{(j)} - 1|. \quad (18)$$

The per-node weighted existence confidence is then

$$\bar{c}_k^{(j)} = \frac{\sum_{i=1}^{M_k} r_{k,i}^{(j)} c_{k,i}^{(j)}}{\epsilon + \sum_{i=1}^{M_k} r_{k,i}^{(j)}}. \quad (19)$$

The final existence-confidence score is mapped into a bounded positive factor:

$$q_{\text{exist},k}^{(j)} = \lambda_{\min} + (1 - \lambda_{\min}) \left(\bar{c}_k^{(j)} \right)^{p_e}, \quad (20)$$

where $\lambda_{\min} \in (0, 1]$ and $p_e > 0$ are tunable parameters. Figure 3 visualizes this bounded Bernoulli-level mapping and representative node-level aggregation effects.

Remark 2 (existence confidence). The mapping $|2r - 1|$ assigns low confidence to an undecided Bernoulli component near $r = 0.5$ and high confidence to decisive absence or presence. Weighting this certainty by $r_{k,i}^{(j)}$ makes the node-level score emphasize components that are likely to exist. The lower bound $\lambda_{\min}$ weakens the influence of an uncertain existence profile without erasing the neighbor. Its spatial information may still be useful.

4.3. Branch-decoupled allocation and FID-FIA refinement

The proposed method allows spatial fusion and existence fusion to use different adaptive scores. This separation follows the Bernoulli/LMB structure because existence and spatial terms enter the fused posterior through different update blocks. Appendix C gives the formal variational interpretation. The method therefore constructs branch-specific dedicated scores:

$$\tilde{\omega}_{k,s,\text{sp}}^{(j)} = m_k^{(j)} \cdot \left(q_{\text{cov},k}^{(j)} \right)^{\alpha_x} \cdot \left(q_{\text{link},k}^{(j)} \right)^{\beta_x}, \quad (21)$$

$$\tilde{\omega}_{k,s,\text{ex}}^{(j)} = m_k^{(j)} \cdot \left(q_{\text{link},k}^{(j)} \right)^{\beta_r} \cdot \left(q_{\text{exist},k}^{(j)} \right)^{\alpha_r}. \quad (22)$$

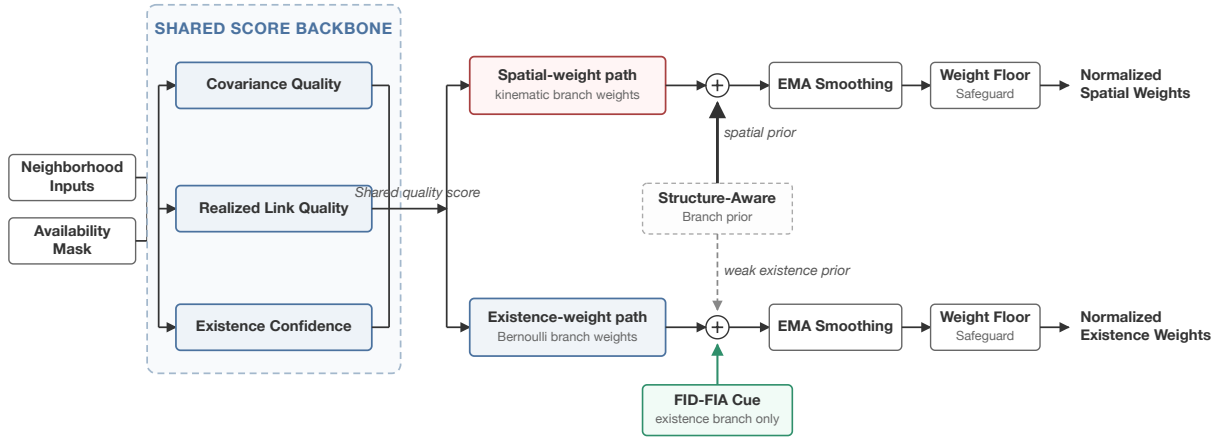


Figure 2: Adaptive fusion-weight factorization, showing the shared quality backbone, decoupled spatial and existence-weight paths, weak structure-aware branch prior, and an FID-FIA cue applied only to the existence branch.

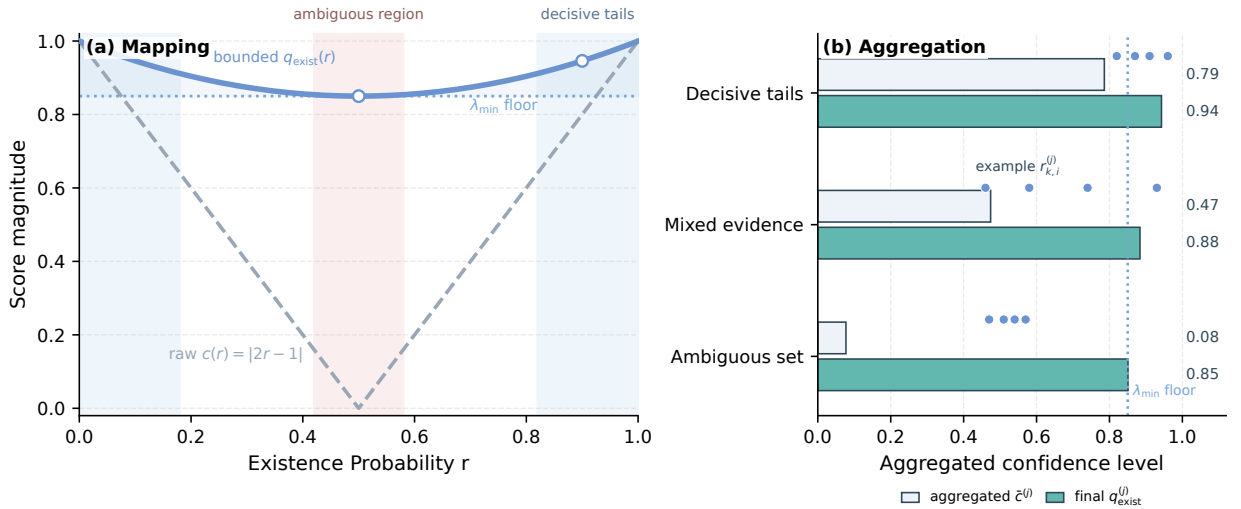


Figure 3: Existence-confidence weighting used in the adaptive rule: (a) the Bernoulli-level decisiveness mapping and bounded score curve, and (b) representative node-level aggregation examples showing how decisive posterior sets produce larger $\tilde{c}^{(j)}$ and $q_{\text{exist}}^{(j)}$.

The powers $\alpha_x, \beta_x, \beta_r, \alpha_r \geq 0$ tune the influence of the branch-relevant cues. The final branch scores are obtained through geometric interpolation:

$$\tilde{\omega}_{k,s}^{x,(j)} = \left(\tilde{\omega}_{k,s}^{(j)} \right)^{1-\eta_x} \left(\tilde{\omega}_{k,s,\text{sp}}^{(j)} \right)^{\eta_x}, \quad \tilde{\omega}_{k,s}^{r,(j)} = \left(\tilde{\omega}_{k,s}^{(j)} \right)^{1-\eta_r} \left(\tilde{\omega}_{k,s,\text{ex}}^{(j)} \right)^{\eta_r}. \quad (23)$$

Here $\eta_x, \eta_r \in [0, 1]$ are interpolation strengths: zero keeps the shared backbone, and one uses the dedicated branch score. This separation connects one reliability backbone to two algebraically different fusion blocks. Spatial and

existence consensus need not respond identically to a weight perturbation. The separation also determines how Fisher information enters the method. Cao and Zhao's construction provides a scalar indicator of target separability and posterior information [17]. In our experiments, this cue is especially effective for cardinality and existence decisions. Related multirate fusion work also supports Fisher information as a local information measure after time alignment [33]. Applying one Fisher-based weight to the entire posterior can degrade spatial RMSE in the present GA-LMB setting. Appendix C shows that this scalar baseline constrains both branches to use the same Fisher utility. Our formulation relaxes that constraint. Covariance and realized link quality continue to govern spatial fusion, whereas FID-FIA is reserved for Bernoulli existence.

Remark 3 (why geometric interpolation). The shared score keeps both branches tied to the same reliability model. The dedicated spatial and existence scores then tilt the anchor toward the evidence most relevant to each branch. Geometric interpolation preserves the multiplicative reliability interpretation: a weak cue continues to constrain the final score even when another term is large.

Weak structure-aware refinement and stabilization. The evaluated Balanced and Cardinality-critical modes next apply a weak graph-aware prior after branch decoupling and before the optional FID-FIA existence modulation. This prior combines local neighborhood overlap with a soft communication-reliability modulation. For fusion node s and neighbor $j \in \mathcal{N}_s$, define the neighborhood-overlap similarity

$$\chi_s^{(j)} = \frac{|\mathcal{N}_s \cap \mathcal{N}_j|}{|\mathcal{N}_s \cup \mathcal{N}_j|}. \quad (24)$$

The spatial branch favors structurally redundant neighbors, while the existence branch gives only a mild novelty preference to neighbors with nonidentical neighborhoods:

$$a_{x,s}^{(j)} = 1 + \chi_s^{(j)}, \quad a_{r,s}^{(j)} = 1 + \frac{1}{2} (1 - \chi_s^{(j)}). \quad (25)$$

Both priors are normalized to unit mean inside the local neighborhood,

$$\xi_{x,s}^{(j)} = \frac{a_{x,s}^{(j)}}{|\mathcal{N}_s|^{-1} \sum_{u \in \mathcal{N}_s} a_{x,s}^{(u)}}, \quad \xi_{r,s}^{(j)} = \frac{a_{r,s}^{(j)}}{|\mathcal{N}_s|^{-1} \sum_{u \in \mathcal{N}_s} a_{r,s}^{(u)}}. \quad (26)$$

When tiered packet-loss statistics are available, the implementation also forms a normalized nominal-reliability prior from the persistent drop tier of each neighbor:

$$b_s^{(j)} = \lambda_c + (1 - \lambda_c) \left(1 - p_{\text{drop}}^{(j)} \right), \quad \zeta_s^{(j)} = \frac{b_s^{(j)}}{|\mathcal{N}_s|^{-1} \sum_{u \in \mathcal{N}_s} b_s^{(u)}}. \quad (27)$$

This slow prior is distinct from the realized link-quality score in the shared backbone. Specifically, $q_{\text{link},k}^{(j)}$ uses delivered and dropped counts from the current step, whereas $\zeta_s^{(j)}$ summarizes the configured or long-run drop tier and does not change with the current packet outcome. The resulting structure factors are

$$\bar{\xi}_{x,s}^{(j)} = \xi_{x,s}^{(j)} \left(\zeta_s^{(j)} \right)^{\delta_c}, \quad \bar{\xi}_{r,s}^{(j)} = \xi_{r,s}^{(j)} \left(\zeta_s^{(j)} \right)^{\delta_c}, \quad (28)$$

and they enter the two branch scores multiplicatively:

$$\tilde{\omega}_{k,s}^{x,(j)} \leftarrow \tilde{\omega}_{k,s}^{x,(j)} \left(\bar{\xi}_{x,s}^{(j)} \right)^{\gamma_x}, \quad \tilde{\omega}_{k,s}^{r,(j)} \leftarrow \tilde{\omega}_{k,s}^{r,(j)} \left(\bar{\xi}_{r,s}^{(j)} \right)^{\gamma_r}. \quad (29)$$

Equivalently, for positive scores, the refinement adds $\gamma_x \log \bar{\xi}_{x,s}^{(j)}$ and $\gamma_r \log \bar{\xi}_{r,s}^{(j)}$ to the spatial and existence log-utilities. The main experiments use $\gamma_x = 0.45$, $\gamma_r = 0.08$, $\delta_c = 0.30$, and $\lambda_c = 0.25$, so the structure prior is deliberately weaker on the existence branch and cannot dominate the covariance, delivery, and existence-confidence backbone.

The raw adaptive weights can fluctuate sharply over time, especially when packet delivery changes abruptly. To improve stability, the normalized weights are smoothed by an exponential moving average and then passed through a

minimum-weight safeguard before final renormalization. This stabilization step preserves the intended quality ordering while preventing unstable weight collapse in a time-varying communication setting.

Remark 4 (non-Gaussian contamination and weight collapse). The experiments use Gaussian measurement errors, and the covariance cue comes from a moment-projected Gaussian mixture. Fallback normalization, temporal smoothing, and a final weight floor protect an active neighbor from a transient zero score. They do not make a deceptively sharp covariance trustworthy. Under impulsive or heavy-tailed contamination, a precise outlier may therefore receive excessive spatial weight. A robust implementation could instead use a shrinkage or M-estimated scatter matrix for $T_{k,i}^{(j)}$. It could also use clipped whitened residuals to form a bounded innovation factor and cap each weight before normalization. These changes preserve the separated simplex allocation, but their thresholds depend on the assumed contamination model. We do not evaluate them here. The reported results therefore provide no evidence of robustness to non-Gaussian noise.

Remark 5 (trust and Byzantine scope). The present communication model treats packet loss and outages as non-adversarial and assumes that posterior parameters and delivery counters reported by active neighbors are honest. A Byzantine node could violate this assumption by advertising an artificially small covariance or a deceptive existence probability. A security-aware extension could use label-aligned spatial residuals and Bernoulli log-odds agreement to form a bounded credibility factor. Authenticated delivery metadata and per-neighbor clipping or trimming could further limit malicious influence before normalization. Resilient labeled-RFS fusion provides relevant mechanisms for this extension [23]. Such a layer is outside the current algorithm; the packet-loss experiments therefore establish communication robustness, not Byzantine resilience.

Existence-only FID-FIA refinement. Cardinality-critical mode adds an information-geometric refinement only to the existence branch. It accumulates pairwise Fisher information distances between locally represented targets and uses the result to modulate existence. For node j , the target-pair score $a_k^{(j)}$ is computed from measurement-updated Bernoulli components. Each component is first projected to one Gaussian. Target pairs are weighted by their existence probabilities. Their distance is then accumulated along the connecting line segment under the sensor's local Fisher metric. The metric depends on detection probability and measurement covariance. Finally, the score is normalized over the available neighborhood,

$$\hat{a}_{k,s}^{(j)} = \frac{a_k^{(j)}}{\epsilon + \max_{u \in \mathcal{N}_s} a_k^{(u)}}, \quad 0 \leq \hat{a}_{k,s}^{(j)} \leq 1. \quad (30)$$

This normalized accumulation is mapped into a bounded existence-only modulation factor,

$$q_{\text{fid},k,s}^{(j)} = \rho_{\min} + (1 - \rho_{\min}) \left(\hat{a}_{k,s}^{(j)} \right)^{p_f}. \quad (31)$$

Here $\rho_{\min} \in [0, 1]$ is the lower bound of the FID-FIA existence cue and $p_f > 0$ controls the nonlinearity of the normalized accumulation. Only the existence score is modified:

$$\tilde{\omega}_{k,s}^{r,(j)} \leftarrow \tilde{\omega}_{k,s}^{r,(j)} \left(q_{\text{fid},k,s}^{(j)} \right)^{\gamma_f}. \quad (32)$$

Here $\gamma_f \geq 0$ controls the strength of the FID-FIA modulation. The spatial score $\tilde{\omega}_{k,s}^{x,(j)}$ is left unchanged. In the main experiment, this branch uses a strong existence-only modulation and disables smoothing and flooring for the existence weights. The spatial branch retains its structure-aware stabilized weights. Equivalently, for positive $q_{\text{fid},k,s}^{(j)}$, the update adds $\gamma_f \log q_{\text{fid},k,s}^{(j)}$ only to the existence-branch utility. The zero-floor setting is the boundary case that permits a zero instantaneous existence weight.

4.4. Algorithm and operating modes

The resulting per-node update is summarized in Algorithm 1. The algorithm uses the quantities consumed by the implemented fusion layer. Spatial weights enter the Gaussian canonical-form update, whereas existence weights enter the Bernoulli existence update.

Algorithm 1. Communication-aware branch-decoupled GA-LMB fusion at node s and time k .

1. Assemble the local neighborhood input from the delivered measurement streams and realized delivery statistics, then obtain the measurement-updated LMB posteriors $\{\pi_k^{(j)} : j \in \mathcal{N}_s\}$ for the current step.
2. Form the availability mask $m_k^{(j)}$, covariance score $q_{\text{cov},k}^{(j)}$, link score $q_{\text{link},k}^{(j)}$, and existence-confidence score $q_{\text{exist},k}^{(j)}$.
3. Build the shared score $\tilde{\omega}_{k,s}^{(j)}$ and the spatial/existence dedicated scores $\tilde{\omega}_{k,s,\text{sp}}^{(j)}$ and $\tilde{\omega}_{k,s,\text{ex}}^{(j)}$.
4. Interpolate the shared and dedicated scores to obtain $\tilde{\omega}_{k,s}^{x,(j)}$ and $\tilde{\omega}_{k,s}^{r,(j)}$, then apply the weak structure-aware branch prior.
5. For Cardinality-critical mode only, compute $q_{\text{fid},k,s}^{(j)}$ and multiply it into the existence score; leave the spatial score unchanged.
6. Normalize each branch over active neighbors, optionally apply temporal smoothing and a final weight floor, and write the results as $\omega_{k,s}^{x,(j)}$ and $\omega_{k,s}^{r,(j)}$.
7. Fuse each aligned Bernoulli component by using $\omega_{k,s}^{x,(j)}$ in the Gaussian canonical-form update and $\omega_{k,s}^{r,(j)}$ in the Bernoulli existence update.

Computational cost. Let $A_{k,s} = |\mathcal{A}_{k,s}|$ be the number of active neighbors and M_k the number of aligned Bernoulli labels. Let $\bar{G}_k$ be the mean mixture size per local Bernoulli component and d_x the state dimension. Forming the covariance backbone requires one moment projection per active neighbor and label, for $O(A_{k,s}M_k\bar{G}_kd_x^2)$ time. The remaining weight operations add $O(A_{k,s}M_k + A_{k,s})$ time. Neighborhood-overlap priors are computed once with the local graph, and their per-update modulation costs $O(A_{k,s})$. Thus Balanced-mode weight formation is linear in both active neighbors and aligned labels for fixed state dimension and mixture size.

For each aligned Bernoulli component, the projected-Gaussian GA merger costs $O(A_{k,s}(\bar{G}_kd_x^2 + d_x^3))$. The first term is moment projection; the second covers covariance solves, inverses, and determinants. The Bernoulli existence product adds only $O(A_{k,s})$. Cardinality-critical mode additionally evaluates all retained target pairs. With Q line-segment quadrature points and C_F the cost of one fixed-dimension Fisher-metric evaluation, its FID-FIA term costs $O(A_{k,s}M_k^2QC_F)$ in the worst case. The implementation uses $Q = 3$. This quadratic pair term, rather than branch normalization, explains the larger measured runtime of FID-FIA-based modes in Section 6.

Operating modes. The method family supports two operating modes. *Balanced mode* uses the three-factor backbone, separate spatial and existence scores, a weak graph prior, and temporal stabilization. It leaves the final FID-FIA modulation inactive. This mode is intended for deployments constrained by localization accuracy, RMSE stability, runtime, or energy.

Cardinality-critical mode adds FID-FIA only to the existence branch. It is intended for settings where target-number errors, missed tracks, or false support outweigh moderate costs in RMSE and runtime. Both modes share the same backbone and branch algebra. They differ only in the final existence cue. The experiments show the resulting tradeoff. Balanced mode favors localization at low cost, whereas Cardinality-critical mode favors cardinality agreement with substantially more computation.

Optional extension modules. The evaluated software also supports several exploratory extension modules. Appendix B summarizes them because their empirical effects are smaller, more coupled, or less stable than the evidence for the main three-factor backbone plus the retained branch-specific refinements.

5. Experimental Setup

This section defines the evaluation protocol used to support the paper's claims. The experiments are organized around one main scenario and a small set of supporting studies. The main scenario is the dual-formation eight-sensor distributed GA-LMB tracking problem under tiered heterogeneous packet loss. Supporting studies check whether the observed gains persist under ideal communication, how the method behaves as communication becomes more constrained, and how optional modules compare with the retained design.

5.1. Scenario, protocols, and operating points

The evaluation follows a hierarchy. The main scenario uses an eight-sensor formation with tiered packet loss and provides the confirmatory evidence. A matched factor ablation identifies which parts of the weight design carry the result. Runtime in the same scenario measures deployment cost. An ideal-communication study tests whether

the refinements remain useful without packet loss. A four-level sensitivity study then examines Balanced mode as communication becomes more constrained. The appendix reports exploratory variants.

Main scenario. The primary evaluation setting uses eight mobile sensors arranged as two four-sensor formations. The filtering mode is distributed local fusion. Each node forms a neighborhood problem from its own and its neighbors' delivered measurement streams. It then obtains measurement-updated LMB posteriors for those streams and fuses the resulting parameters. The local multi-object filter is GA-LMB with loopy belief propagation (LBP) for approximate marginal data association [38].

The choice of the geometric-average KLA route is deliberate. GA fusion gives a conservative logarithmic-opinion-pool update under unknown cross-correlations. It also admits a stable canonical-form update for LMB spatial densities and exposes the Bernoulli product rule needed for branch-specific weighting. These properties make it a cleaner vehicle for studying adaptive weight allocation than a broad comparison across fusion families.

The common sensor-side configuration is:

- number of sensors: 8,
- filter mode: GA-LMB,
- data association: LBP,
- clutter rate: 3 for every sensor,
- detection probability: 0.9 for every sensor,
- measurement-noise scale: $q = 3$ for every sensor,
- sensor field of view enabled with half-angle 60° and range 60000.

The nominal detection probability, measurement-noise scale, and clutter rate are therefore homogeneous across sensors. The main sensing variation comes from the dual-formation geometry, finite fields of view, and target visibility. These conditions induce differences in posterior covariance and existence, but they do not vary explicit p_D , measurement-noise, or clutter tiers.

The target side consists of ten objects arranged in three groups with staggered births. The three groups contain $3 + 3 + 4$ targets, respectively, and the targets move toward the central region with nominal speed 0.45. Births are staggered every eight time steps, and the total simulation length is 100 steps.

Communication model. The main scenario uses a tiered heterogeneous packet-loss setting. Communication is applied before distributed fusion through a measurement-stream delivery model with bandwidth limitation and per-sensor packet loss. The main configuration is:

- communication level: 2,
- global maximum delivered measurements per step: 80,
- packet-selection policy: weighted priority with random measurement selection,
- link model: fixed packet drop,
- target mean drop rate: $p_{\text{drop}} = 0.2$,
- tiered drop levels: $[0, 0.1, 0.2, 0.5]$,
- tiered level counts: $[1, 4, 1, 2]$.

This tiered design preserves the historical mean communication degradation while introducing persistent cross-node heterogeneity.

Figure 4 summarizes the contrast between the simpler standard fixed ideal benchmark and the main tiered-loss scenario. The standard ideal benchmark uses four full region-of-interest (ROI) sensors, ideal communication, and fixed target births. The main scenario is the dual-formation eight-sensor case with staggered target waves, a bandwidth cap, and persistent drop-tier heterogeneity before distributed fusion.

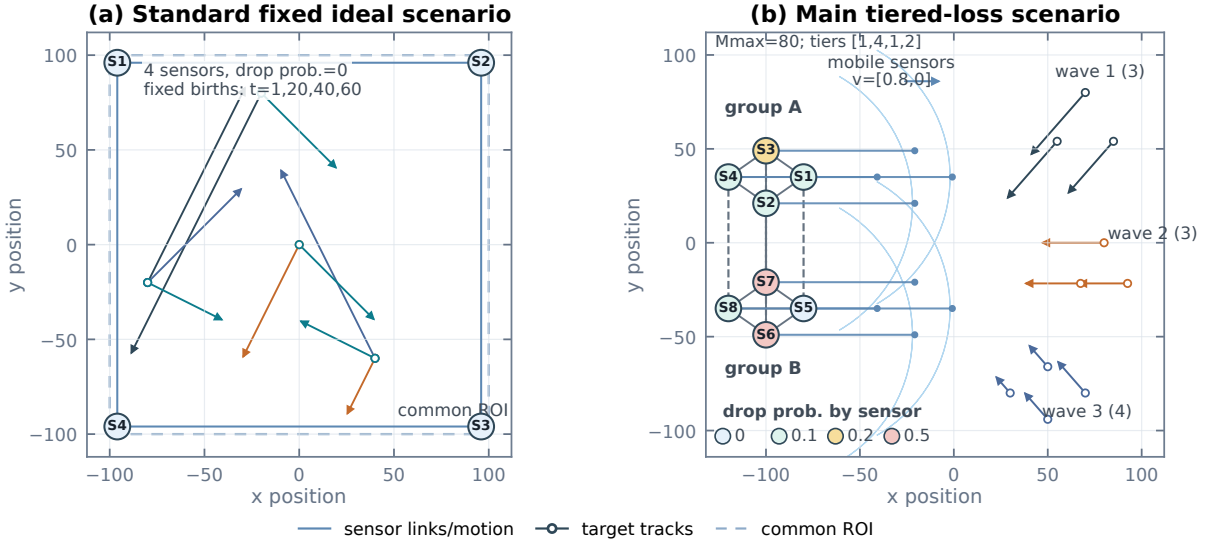


Figure 4: Schematic comparison between the standard fixed ideal benchmark and the main tiered heterogeneous packet-loss scenario. The ideal benchmark uses four full-ROI sensors with no packet loss and fixed target births; the main scenario uses the dual-formation eight-sensor geometry and staggered target waves with the bandwidth cap and persistent drop-tier heterogeneity used in the main experiments.

Mainline ablation protocol. The core ablation study compares five consistently named arms:

1. Fixed Metropolis,
2. Covariance-only adaptive,
3. Covariance-link adaptive,
4. Three-factor adaptive backbone,
5. Balanced mode, which adds weak structure-aware modulation after covariance, link-quality, and existence-confidence weighting.

This five-arm path diagnoses the backbone before the final existence refinement. The results table focuses on operating points and treats the three-factor backbone as part of Balanced mode. The main comparison also includes probability-of-detection (PD)-weighted GA, scalar FID-FIA-weighted GA, and Cardinality-critical mode. Balanced mode uses one parameterization across all trials. Exponential smoothing is $\alpha_{\text{EMA}} = 0.7$, and the minimum weight is 0.05. Existence-confidence parameters are $\lambda_{\min} = 0.85$ and $p_e = 2.0$. Spatial and existence interpolation strengths are 0.5 and 0.15, and the graph-prior strengths are 0.45 and 0.08. The nominal communication prior uses power 0.30 and lower bound 0.25. Cardinality-critical mode leaves the spatial branch unchanged but adds FID-FIA to existence. Smoothing and minimum-weight flooring are disabled for this final existence refinement. Other exploratory modules are inactive.

Reviewer-requested external-baseline adaptations. Two additional rows map the closest peer-to-peer methods onto the same one-round GA-LMB protocol [23, 28]. We label both rows as adaptations because their source methods use a different filter implementation or threat model. Zheng et al. formulate weighted RFS subdensity fusion and provide consensus and flooding implementations for GM-PHD filters [28]. They do not provide an LMB implementation. In our adaptation, each label-associated Bernoulli component is one subdensity. For label ℓ and active neighbor j , the spatial and existence coefficients are

$$a_{x,j}^{(\ell)} = \frac{1}{\text{tr}(P_{j,\text{pos}}^{(\ell)}) + \epsilon}, \quad a_{r,j}^{(\ell)} = \epsilon_r + (1 - \epsilon_r) |2r_j^{(\ell)} - 1|, \quad (33)$$

where ϵ is machine precision and $\epsilon_r = 0.05$. Each coefficient vector is normalized over available neighbors and applied to the corresponding GA-LMB branch. The adaptation retains subdensity-level weighting and separate localization and cardinality cues. It is not equivalent to the source GM-PHD feedback or flooding implementations.

Gao et al. address corrupted LMB messages and adversarial nodes [23]. Each receiving node treats its local density as trusted, detects modified or false Bernoulli components, and reduces confidence as discrepancy grows. Our adaptation aligns components by label and assigns neighbor j the credibility

$$c_{s,j}^{(\ell)} = \max \left\{ \epsilon_c, \exp \left[-\frac{J_{G,s,j}^{(\ell)}/d_x + J_{B,s,j}^{(\ell)}}{\tau} \right] \right\}, \quad \omega_{s,j}^{(\ell)} \propto \bar{\omega}_{s,j} c_{s,j}^{(\ell)}, \quad (34)$$

Here, J_G and J_B are symmetric KL divergences for the projected Gaussian and Bernoulli distributions. We set $d_x = 4$, $\tau = 2$, and $\epsilon_c = 0.05$. The coefficient $\bar{\omega}_{s,j}$ is the Metropolis weight, and the receiving node has unit credibility. Hard rejection is disabled because the experiment modifies packet delivery but not posterior parameters. This row therefore evaluates soft local-trust weighting, not the full attack detector. Every adaptation and proposed method uses the same truth, measurements, delivery realization, and 50 deterministic paired seeds 2–51. The adaptation arms and the reused core arms are executed in separate run cohorts. Their accuracy metrics remain paired by seed, but we do not report cross-cohort relative runtime.

5.2. Metrics and reporting

The experiments report three categories of metrics. The first category is network-level disagreement among sensors after distributed fusion. These primary metrics are OSPA consensus error, GOSPA consensus error, matched localization disagreement, and cardinality dispersion as defined in Section 3. The second category is conventional truth-referenced tracking accuracy. These secondary metrics are local Euclidean OSPA (E-OSPA), local root mean square error (RMSE), and local cardinality error. The third category is computational cost, measured as filter/fusion wall-clock runtime per trial, runtime per simulation step, and runtime relative to Fixed Metropolis. Every Euclidean OSPA and GOSPA calculation uses cutoff $c = 5$ and order $p = 2$, with $\alpha = 2$ for GOSPA. Their ground distance uses the same complete extracted kinematic state vectors. OSPA and GOSPA provide complementary normalized and unnormalized finite-set distances for multi-object filtering and tracking evaluation [34–36]. The primary disagreement metrics are task-specific network-agreement measures built from these standard ingredients, and the paper interprets them together with the secondary local metrics and computational-cost measurements.

This hierarchy reflects the method's purpose. The method is not a generic single-sensor accuracy booster. It reduces disagreement in the fused multisensor picture under heterogeneous communication. Network disagreement is therefore the main outcome. Local tracking metrics guard against hidden accuracy losses, and runtime measures deployment cost. The target is better agreement without a meaningful loss in conventional tracking quality or impractical computation. Runtime covers only the distributed LMB filtering and fusion call for each arm. Scenario generation, communication sampling, plotting, and metric evaluation are excluded.

Unless otherwise stated, reported scalar metrics are Monte Carlo means. The main fixed-versus-existence-refined confirmation, the five-arm factor ablation, the ideal-communication comparison, and the communication-level sensitivity study all use 50 deterministic paired trials with seeds 2–51. The main confirmation, ideal-communication comparison, and communication-level sensitivity study are paired by construction. The five-arm factor ablation remains primarily a design diagnostic, but it is now reported on the same 50-trial seed set for consistency.

6. Results and Discussion

This section reports the empirical evidence supporting the paper's main claims. The strongest evidence comes from the tiered heterogeneous packet-loss GA scenario introduced in Section 5, so that scenario is treated as the primary result. Ideal communication and communication-sensitivity experiments are used as supporting studies. Appendix-only results document exploratory variants.

6.1. Main comparison and design ablation

Consensus and local safeguards. We first tested whether communication-aware weighting improved agreement in the eight-sensor GA-LMB scenario with tiered packet loss. Network disagreement is reported before truth-referenced

Table 1

Main paired consensus comparison under tiered heterogeneous packet loss. OSPA and GOSPA use $(c, p) = (5, 2)$ on the complete extracted state, with $\alpha = 2$ for GOSPA. Values are mean $\pm$ standard deviation over 50 deterministic paired trials with seeds 2–51. Lower values are better.

Arm	OSPA err.	GOSPA err.	Loc. disag.	Card. disp.
Fixed Metropolis	2.469 ± 0.220	4.780 ± 0.245	2.326 ± 0.353	0.716 ± 0.224
PD-weighted GA	2.177 ± 0.161	4.306 ± 0.191	1.995 ± 0.233	0.588 ± 0.170
FID-FIA-weighted GA	1.818 ± 0.056	4.045 ± 0.107	1.643 ± 0.109	0.123 ± 0.023
Balanced mode	1.779 ± 0.073	3.909 ± 0.124	1.522 ± 0.157	0.188 ± 0.031
Cardinality-critical mode	1.678 ± 0.050	3.904 ± 0.114	1.546 ± 0.158	0.063 ± 0.017

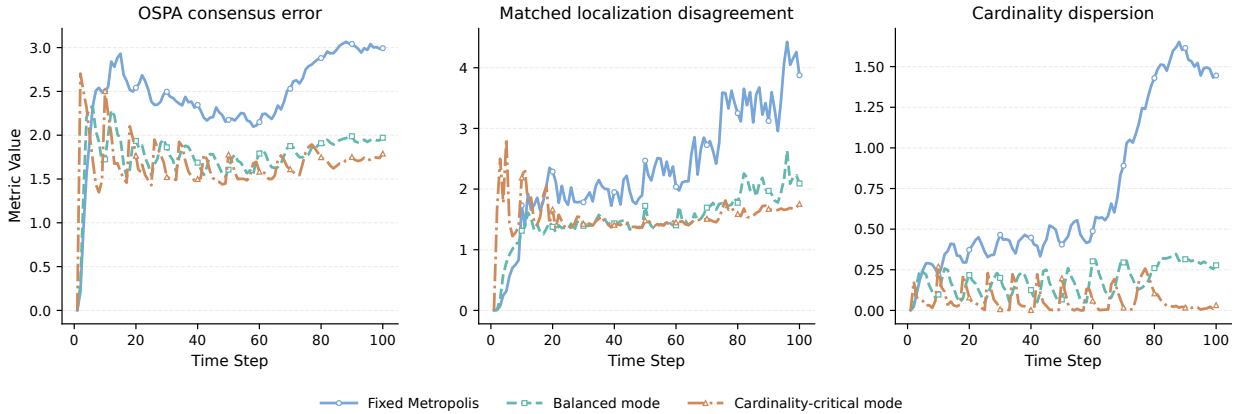


Figure 5: Mean OSPA consensus error, matched localization disagreement, and cardinality dispersion over time for Fixed Metropolis, Balanced mode, and Cardinality-critical mode under tiered heterogeneous packet loss. Curves are 50-trial means over deterministic paired trials with seeds 2–51; distinct line styles and sparse markers at ten-step intervals separate the trajectories without obscuring the temporal trend.

local accuracy because agreement is the method’s primary target. The comparison addresses three questions. First, can a direct reliability weight improve GA fusion? Second, does separation of the spatial and existence branches add value? Third, does an existence-specific Fisher cue help when cardinality errors dominate? The probability-of-detection (PD)-weighted GA row addresses the first question by weighting each target with its local detection probability.

Table 1 shows the progression across method variants. PD-weighted GA improved on Fixed Metropolis, so direct reliability weighting was already useful under heterogeneous delivery. Scalar FID-FIA weighting was particularly effective for cardinality dispersion. The proposed modes gave the stronger overall balance. Balanced mode achieved the lowest localization disagreement. Cardinality-critical mode achieved the lowest OSPA and GOSPA consensus errors and cardinality dispersion. Relative to Fixed Metropolis, its OSPA, localization, and cardinality metrics fell from 2.469 to 1.678, from 2.326 to 1.546, and from 0.716 to 0.063. These changes correspond to reductions of 32.1%, 33.5%, and 91.2%. Its unnormalized GOSPA error fell from 4.780 to 3.904, an 18.3% reduction. For each proposed mode, all 50 paired trials improved on Fixed Metropolis for all four network metrics. The exact two-sided sign-test value was $p = 1.8 \times 10^{-15}$ in each comparison. Figure 5 shows unsmoothed, 50-trial means of three temporal trajectories. Both modes remained below Fixed Metropolis for most time steps, with the clearest cardinality separation in Cardinality-critical mode.

Table 2 checks whether the consensus gains conceal a loss in local tracking accuracy. PD-weighted GA achieved the lowest truth-referenced RMSE, but it was weaker in E-OSPA and local cardinality error. Cardinality-critical mode showed the complementary pattern. It achieved the best E-OSPA and cardinality error, but its RMSE exceeded those of PD-weighted GA, Fixed Metropolis, and Balanced mode. Its RMSE nevertheless remained slightly below that of scalar FID-FIA weighting. The reduction in cardinality dispersion therefore coincided with better target-count estimates. This pattern is consistent with applying Fisher information only to Bernoulli existence while retaining the communication-aware spatial branch.

Table 2

Conventional local tracking metrics in the main tiered heterogeneous packet-loss scenario. Values are mean $\pm$ standard deviation across the same 50 deterministic paired trials used in Table 1, after averaging each trial over sensors.

Arm	E-OSPA	RMSE	CardErr
Fixed Metropolis	2.863 $\pm$ 0.124	1.650 $\pm$ 0.078	1.455 $\pm$ 0.242
PD-weighted GA	2.736 $\pm$ 0.116	1.563 $\pm$ 0.061	1.255 $\pm$ 0.205
FID-FIA-weighted GA	2.185 $\pm$ 0.050	1.734 $\pm$ 0.094	0.388 $\pm$ 0.047
Balanced mode	2.335 $\pm$ 0.072	1.606 $\pm$ 0.050	0.579 $\pm$ 0.067
Cardinality-critical mode	2.020 $\pm$ 0.047	1.721 $\pm$ 0.161	0.224 $\pm$ 0.029

Table 3

Comparison with reviewer-requested literature adaptations in the main tiered-loss scenario. Values are mean $\pm$ standard deviation over 50 deterministic paired seeds. The Fixed and proposed rows reuse the validated core cohort; the two adaptation rows use the same seeds and communication realizations in a separate cohort. Consequently, accuracy is paired by seed, but cross-cohort relative runtime is not reported. Lower values are better.

(a) Network disagreement				
Arm	OSPA err.	GOSPA err.	Loc. disag.	Card. disp.
Fixed Metropolis	2.469 $\pm$ 0.220	4.780 $\pm$ 0.245	2.326 $\pm$ 0.353	0.716 $\pm$ 0.224
Zheng-style adaptation	3.577 $\pm$ 0.241	5.008 $\pm$ 0.357	13.104 $\pm$ 3.144	0.969 $\pm$ 0.226
Gao-style adaptation	3.332 $\pm$ 0.149	6.203 $\pm$ 0.152	4.048 $\pm$ 0.581	1.438 $\pm$ 0.210
Balanced mode	1.779 $\pm$ 0.073	3.909 $\pm$ 0.124	1.522 $\pm$ 0.157	0.188 $\pm$ 0.031
Cardinality-critical mode	1.678 $\pm$ 0.050	3.904 $\pm$ 0.114	1.546 $\pm$ 0.158	0.063 $\pm$ 0.017

(b) Truth-referenced local safeguards			
Arm	E-OSPA	RMSE	CardErr
Fixed Metropolis	2.863 $\pm$ 0.124	1.650 $\pm$ 0.078	1.455 $\pm$ 0.242
Zheng-style adaptation	4.243 $\pm$ 0.244	2.906 $\pm$ 0.732	4.202 $\pm$ 0.550
Gao-style adaptation	3.092 $\pm$ 0.108	2.184 $\pm$ 0.238	2.002 $\pm$ 0.247
Balanced mode	2.335 $\pm$ 0.072	1.606 $\pm$ 0.050	0.579 $\pm$ 0.067
Cardinality-critical mode	2.020 $\pm$ 0.047	1.721 $\pm$ 0.161	0.224 $\pm$ 0.029

The two proposed operating-mode rows are recommended settings of the same method family. The Balanced mode is preferable when spatial RMSE stability is the dominant requirement. The Cardinality-critical mode is preferable when target-number agreement and truth-referenced cardinality error are the dominant risks. This usage-oriented distinction makes the tradeoff explicit.

Reviewer-requested literature adaptations. We next tested whether the gains remained under the two literature-motivated adaptations defined in Section 5. The Zheng-style row preserves subdensity-level branch weighting but maps the source GM-PHD consensus/flooding formulation to label-associated Bernoulli components in our one-round GA-LMB protocol. The Gao-style row preserves receiver-local trust and distance-based attenuation. Its hard rejection is disabled because this experiment contains packet loss but no corrupted posterior parameters. These rows therefore test the portability of the published weighting principles, not exact source-algorithm reproductions.

Table 3 shows that neither adaptation transferred favorably to this communication-limited one-round protocol. Balanced mode had lower OSPA, GOSPA, localization disagreement, and cardinality dispersion than each adaptation in all 50 paired trials. Cardinality-critical mode improved on both adaptations in all paired comparisons except GOSPA against the Zheng-style adaptation, where it improved in 49 of 50 trials. The exact two-sided sign-test values were $p = 1.8 \times 10^{-15}$ for the 50-of-50 comparisons and $p = 9.1 \times 10^{-14}$ for the 49-of-50 comparison. The lower local E-OSPA, RMSE, and cardinality-error means in the proposed rows indicate that the network agreement was not obtained by forcing inaccurate local estimates into consensus.

The failure modes are informative. The Zheng-style adaptation uses posterior concentration and existence margin but no realized delivery-quality cue. Under partial FOV and packet loss, a concentrated component can still be poorly aligned with complementary neighbors, which coincided with high localization disagreement. The Gao-style adaptation measures each neighbor against the receiver's local posterior. In an honest network with legitimate cross-view differences, this local-trust prior can attenuate useful complementary evidence even though no attack is present.

Table 4

Computational cost in the main tiered heterogeneous packet-loss scenario. Wall-clock time measures only the distributed LMB filtering/fusion call for each arm over 50 deterministic paired trials with seeds 2–51. The Fixed Metropolis, FID-FIA-weighted GA, Balanced, and Cardinality-critical rows come from the operating-mode runtime probe; the PD-weighted GA row comes from the direct dynamic-weighting probe under the same scenario and seed set. Relative runtime is computed against the paired Fixed Metropolis denominator within the corresponding probe.

Arm	Runtime (s)	Runtime/step (s)	Relative runtime
Fixed Metropolis	52.123 ± 7.932	0.521	1.000×
PD-weighted GA	61.668 ± 5.376	0.617	1.233×
FID-FIA-weighted GA	147.674 ± 23.957	1.477	2.833×
Balanced mode	56.378 ± 9.626	0.564	1.082×
Cardinality-critical mode	155.913 ± 18.220	1.559	2.991×

Table 5

Network disagreement metrics for the main factor ablation under tiered heterogeneous packet loss. Values are mean $\pm$ standard deviation over 50 deterministic paired trials. The rows show the progression from fixed weighting to the covariance-link backbone and then to the Balanced operating mode, which includes the three-factor adaptive backbone and weak structure-aware branch decoupling. The final row applies the FID-FIA cue only to the existence-weight path to obtain the cardinality-critical operating mode.

Arm	OSPA err.	Loc. disag.	Card. disp.
Fixed Metropolis	2.469 ± 0.220	2.326 ± 0.353	0.716 ± 0.224
Covariance-only adaptive	2.091 ± 0.145	1.951 ± 0.253	0.466 ± 0.134
Covariance-link adaptive	1.788 ± 0.071	1.525 ± 0.156	0.188 ± 0.031
Balanced mode	1.779 ± 0.073	1.522 ± 0.157	0.188 ± 0.031
Cardinality-critical mode	1.678 ± 0.050	1.546 ± 0.158	0.063 ± 0.017

The resulting cardinality dispersion and GOSPA were high. These findings are specific to the stated adaptations and protocol. They do not establish superiority over the source methods in their native GM-PHD consensus/flooding or adversarial-LMB settings.

Computational cost is the third evaluation axis for this method family. Table 4 reports 50-trial runtime measurements in the same 100-step tiered-drop scenario and covers all arms in the main comparison. The measurement includes only the distributed LMB filtering/fusion call for each arm; scenario generation, communication-model sampling, plotting, and metric evaluation are excluded. The main cost distinction is clear. The Balanced mode remains close to the fixed-weight baseline, requiring about 1.08× the Fixed Metropolis runtime. The PD-weighted GA baseline is also a moderate-cost dynamic-weighting alternative at about 1.23× fixed runtime. By contrast, FID-FIA-weighted GA and the Cardinality-critical mode are both near the three-times-cost regime. The runtime evidence identifies FID-FIA scoring as the dominant source of additional cost.

The cost results sharpen the operating-mode recommendation. The Balanced mode is the low-overhead default when spatial consensus, RMSE stability, runtime, or energy budget is the limiting requirement. The Cardinality-critical mode should be selected when target-number agreement and suppression of missed or false support justify roughly three times the fixed-weight filtering/fusion time.

Factor ablation. The most important diagnostic ablation is the factor-by-factor path from Fixed Metropolis to the Balanced mode. Table 5 summarizes this 50-trial path and adds the Cardinality-critical mode as the last row. The three-factor adaptive backbone is omitted as a separate row to keep the table focused on the main operating points. It remains the reliability model used by the branch-decoupled Balanced mode. The last row is the branch-specific existence refinement that converts the Balanced mode into the cardinality-sensitive mode.

Several conclusions follow from Table 5. First, covariance weighting is necessary but not sufficient. It improves over fixed topology weights and confirms posterior concentration as a useful first-order proxy for local posterior quality. A substantial gap to the branch-decoupled modes nevertheless remains.

Second, realized link quality is the dominant communication-aware factor under heterogeneous packet loss. Adding it after covariance produces the largest incremental change in the backbone path, especially for cardinality dispersion, which is approximately halved. This is the key evidence that the method is not merely reweighting sharper posteriors; it is also responding to whether those posteriors are reliably delivered to neighboring nodes.

Third, existence confidence remains part of the adaptive backbone, and its effect is folded into the branch-decoupled Balanced mode in Table 5. This is a methodological choice: the reliability model is applied differently to the spatial and Bernoulli existence paths. This is consistent with the design rationale, because covariance and link quality do not directly express how decisive a local Bernoulli component is about target existence.

Finally, the Balanced mode provides the pre-FID-FIA branch-decoupled operating point. The final row applies the FID-FIA cue only to the existence branch. Cardinality dispersion decreases from about 0.188 to 0.063, and OSPA consensus error also decreases. This is why the proposed method is presented as a configurable branch-decoupled family: the Balanced mode is the spatial/position-oriented setting, and the Cardinality-critical mode is the cardinality-oriented setting.

6.2. Supporting communication studies

Ideal communication. An obvious question is whether the weak structure-aware decoupled refinement only helps by compensating for packet loss. We therefore set the communication level to zero and compare all five main arms under otherwise matched conditions. Table 6 reports this comparison over the same 50 deterministic seed set used in the main confirmation.

Table 6

Ideal-communication supporting comparison over 50 deterministic paired trials with seeds 2–51. Adaptive rows report the mean value and relative change versus Fixed Metropolis; lower values are better, so negative percentages indicate improvements.

Arm	Network disagreement			Local safeguards	
	OSPA err.	Loc. disag.	Card. disp.	E-OSPA	RMSE
Fixed Metropolis	1.634	1.381	0.090	1.908	1.442
PD-weighted GA	1.434 (-12.2%)	1.218 (-11.8%)	0.056 (-37.6%)	1.825 (-4.3%)	1.377 (-4.5%)
FID-FIA-weighted GA	1.534 (-6.1%)	1.333 (-3.5%)	0.057 (-36.3%)	1.881 (-1.4%)	1.500 (+4.0%)
Balanced mode	1.427 (-12.7%)	1.202 (-13.0%)	0.071 (-20.9%)	1.839 (-3.6%)	1.375 (-4.7%)
Cardinality-critical mode	1.433 (-12.3%)	1.303 (-5.7%)	0.049 (-46.0%)	1.766 (-7.4%)	1.470 (+2.0%)

The ideal-communication result mirrors the branch specialization observed in the tiered-drop experiment. The PD-weighted GA row confirms that a simple reliability-weighted GA baseline can also help when link degradation is removed. Information-geometric weighting is especially effective for cardinality dispersion. The localization-oriented Balanced mode gives the lowest OSPA consensus error, matched localization disagreement, and local RMSE in this benign setting. The Cardinality-critical mode gives the lowest cardinality dispersion and local E-OSPA without forcing a single FID-FIA weight onto the whole posterior. Thus, the refinement is not merely repairing packet loss, and its consensus gain does not coincide with collapsed local tracking quality.

Communication sensitivity and robustness. We next tested Balanced mode across four communication levels. Each level used the same eight-sensor scenario and 50 deterministic paired trials. Level 0 used ideal communication. Level 1 added only the global bandwidth cap, and level 2 added the tiered link-loss model. Level 3 further added random node outages. Every level used seeds 2–51. The level-2 Fixed and Balanced trials are exactly those summarized in Table 1. Table 7 reports the disagreement metrics, and Figure 6 adds 95% confidence intervals. This study varies communication only; it does not vary sensor detection probability or measurement noise.

Balanced mode improved all three disagreement metrics at every communication level. OSPA and localization improved in all 50 trials at each level. Cardinality dispersion improved in 49 of 50 trials at levels 0 and 1, and in all trials at levels 2 and 3. The exact sign-test values were at most 9.1×10^{-14} . Gains were modest under ideal communication or the bandwidth cap alone. They became much larger when link loss or outages created heterogeneous realized delivery. Communication-aware weighting was therefore most useful when link outcomes differed across nodes. We restricted this study to two arms to preserve one coherent protocol. Sensitivity of the more expensive FID-FIA refinement beyond levels 0 and 2 remains future work.

Table 7

Communication-level sensitivity in the eight-sensor dual-formation GA-LMB scenario. Each level uses 50 deterministic paired trials with seeds 2–51. Values are Fixed Metropolis / Balanced mode, and lower values are better. The level-2 entries use the same trials as the corresponding rows of Table 1.

Level	Additional constraint	OSPA err.	Loc. disag.	Card. disp.
0	none	1.634/1.427	1.381/1.202	0.090/0.071
1	bandwidth cap	1.717/1.484	1.451/1.242	0.137/0.086
2	tiered link loss	2.469/1.779	2.326/1.522	0.716/0.188
3	node outage	2.671/1.800	2.725/1.570	0.945/0.193

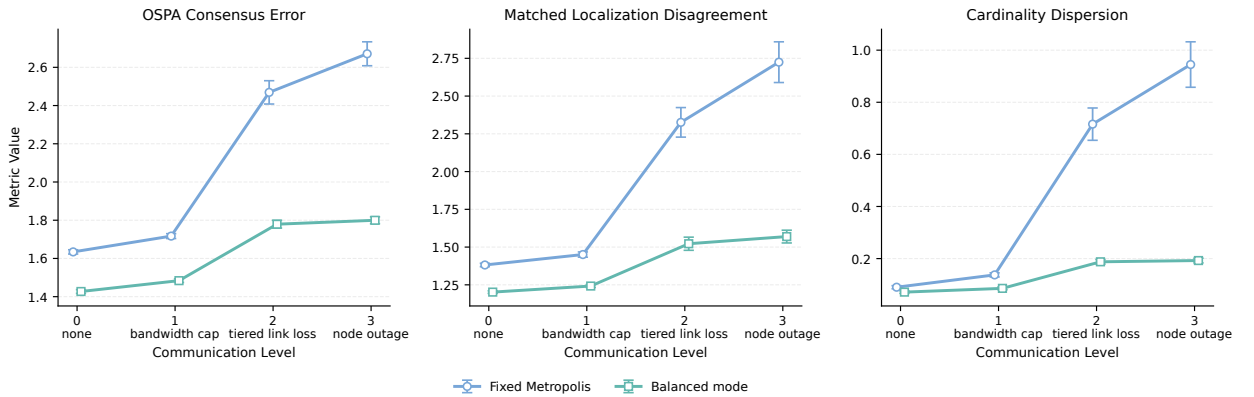


Figure 6: Communication-level sensitivity for Fixed Metropolis and Balanced mode under one coherent paired protocol. Markers show 50-trial means and error bars show 95% confidence intervals.

7. Conclusion

This paper studies communication-aware weights for distributed GA/KLA fusion of LMB posteriors under heterogeneous packet delivery. Fixed topology weights cannot respond when posterior concentration, Bernoulli existence confidence, and realized link quality differ across nodes. The proposed three-factor backbone adapts the fusion weights at low cost. Separate spatial and existence branches allow each part of the LMB posterior to use the relevant reliability cues.

Both proposed modes improved network disagreement in the eight-sensor tiered-loss scenario. Balanced mode kept runtime close to fixed weighting and gave the strongest localization result. Cardinality-critical mode applied FID-FIA only to existence and gave the strongest target-number agreement. Its runtime was comparable to other FID-FIA methods. Both modes also outperformed two explicitly labeled literature-method adaptations on paired network metrics. The ideal-communication study showed that these effects were not limited to packet-loss compensation. Across four communication levels, Balanced mode produced its largest gains when realized delivery became heterogeneous.

The evidence has clear limits. Geometry and FOV vary across sensors, and communication is heterogeneous. Detection probability, clutter rate, and measurement-noise scale remain homogeneous. The experiments also assume honest nodes and Gaussian measurement errors. The literature rows adapt weighting principles to our one-round GA-LMB interface; they are not exact reproductions of the source GM-PHD propagation scheme or adversarial LMB detector. Robustness to packet loss must therefore not be interpreted as robustness to impulsive noise or Byzantine reports. Future work should examine robust scatter estimates, bounded innovation scores, and credibility limits under explicit contamination and attack models. Other priorities include native-protocol comparisons, richer sensing heterogeneity, larger Monte Carlo studies, cheaper FID-FIA implementations, and asynchronous or multirate LMB fusion.

A. Simulation Setup

Section 5 defines the common filter, sensor, communication, operating-mode, and Monte Carlo settings. The remaining constants needed to reproduce the scenario are the formation geometry, target initialization, and paired-run handling below.

The eight sensors form two four-sensor groups. Their formation centers start at $[-80, 35]^T$ and $[-80, -35]^T$ and move with nominal velocity $[0.8, 0]^T$. Each group follows a four-agent leader-follower pattern with spacing 20. The communication graph is fully connected inside each group, with cross links $1 \leftrightarrow 5$, $2 \leftrightarrow 6$, $3 \leftrightarrow 7$, and $4 \leftrightarrow 8$; the communication range is 150.

The ten targets are initialized as the three staggered groups summarized in Section 5. Their group centers are $[70, 80]^T$, $[80, 0]^T$, and $[70, -80]^T$, with spacings 30, 25, and 20, respectively. Births start at time step one, repeat every eight steps, and last through the 100-step scenario.

Within each deterministic seed, all compared arms share the same generated truth, measurements, communication realization, and delivered measurement sets. The ideal-communication comparison reuses the same geometry and targets but sets communication level 0, uses effectively unlimited delivery, and sets $p_{\text{drop}} = 0$. The appendix-only AA diagnostics use the same communication-constrained scenario, with a shorter AA horizon and stronger AA-specific pruning to control mixture growth.

B. Additional Attempted Modules and Results

The secondary experiments below check how far the retained reliability cues transfer beyond the main GA/KLA operating modes. Unless noted, they reuse the tiered-loss scenario and reporting protocol from Section 5.

AA-based secondary route. An arithmetic-average (AA) diagnostic checks whether the same reliability cues transfer to a different fusion operator. The AA run uses the horizon and pruning changes noted in Appendix A. Its merger uses linear Bernoulli averaging and existence-weighted Gaussian-mixture concatenation, rather than GA/KLA logarithmic pooling.

Table 8

Appendix-only AA-based secondary route.

Arm	OSPA	Loc. disag.	Cardinality
fixed AA	3.903	5.130	0.226
covariance-link AA	3.507	4.401	0.103
Balanced AA	3.449	4.306	0.102
Cardinality-critical AA	3.457	4.328	0.095

Balanced AA reduces OSPA consensus error, matched localization disagreement, and cardinality dispersion by about 11.6%, 16.1%, and 55.0% relative to fixed AA. Cardinality-critical AA gives the lowest cardinality dispersion, about a 58.0% reduction, but is slightly worse than Balanced AA on OSPA and localization disagreement. The pattern matches the main operating-mode tradeoff, while the evidence remains secondary because the fusion operator is different.

Other exploratory modules. Several exploratory modules produced weak, mixed, or negative effects in the current experiments:

Table 9

Exploratory modules not retained in the main method.

Module	Observed effect	Decision
Freshness and history weighting	Freshness is effectively flat; history gives only a small localization change while OSPA and cardinality are unchanged or slightly worse.	No main-text role.
Direct cardinality-consensus score	Pilot degrades OSPA, RMSE, and cardinality dispersion relative to the covariance-link baseline.	Negative appendix evidence.
Association ambiguity / stronger posterior structure	Association ambiguity is mixed; stronger posterior-structure consistency ties the retained weak static structure prior.	Future extension candidate.

These results keep the design space visible without turning every tested cue into a method component. The retained method stays focused on covariance quality, realized link quality, existence confidence, branch decoupling, and the existence-branch FID-FIA cue.

C. Theoretical Interpretation of Adaptive Branch-Decoupled Fusion

This appendix interprets the adaptive branch-decoupled rule under the assumptions used by the implementation. Labels are aligned before fusion, and each Gaussian mixture is replaced by its moment-compatible Gaussian projection. The remaining steps use the standard geometric-average KLA and LMB constructions [5–7, 12–15]. The results below therefore explain the implemented local update; they do not characterize unconstrained LMB fusion.

C.1. Entropy-Regularized Interpretation of the Weight Backbone

Fix a fusion node s at time k with active neighborhood $\mathcal{A}_{k,s}$. The adaptive backbone constructs the positive score

$$s_{0,j} = q_{\text{cov},k}^{(j)} q_{\text{link},k}^{(j)} q_{\text{exist},k}^{(j)}, \quad j \in \mathcal{A}_{k,s}, \quad (35)$$

where the factors denote covariance quality, realized link quality, and existence confidence. The optimization below applies to positive scores in the simplex interior. A zero factor is interpreted as a boundary limit. If all active scores vanish, the implementation uses uniform weights over active neighbors. Define the additive utility $u_{0,j} = \log s_{0,j}$. The feasible weight set is

$$\Delta_{k,s} = \left\{ w \in \mathbb{R}^{|\mathcal{A}_{k,s}|} : w_j \geq 0, \sum_j w_j = 1 \right\}. \quad (36)$$

Proposition C1. For any $\tau > 0$, the program

$$\max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j u_{0,j} + \tau H(w) \right\}, \quad H(w) = - \sum_j w_j \log w_j, \quad (37)$$

has the softmax solution

$$w_j^\star = \frac{\exp(u_{0,j}/\tau)}{\sum_u \exp(u_{0,u}/\tau)} = \frac{s_{0,j}^{1/\tau}}{\sum_u s_{0,u}^{1/\tau}}. \quad (38)$$

Setting $\tau = 1$ recovers the implemented normalized product rule. Multiplication combines the reliability factors, and the logarithm makes their utilities additive. The entropy term prevents a degenerate allocation within the simplex interior.

Covariance score in the isotropic limit. If the local Gaussian components satisfy $T_{k,i}^{(j)} = \sigma_j^2 I_d$, then the implemented covariance score

$$q_{\text{cov},k}^{(j)} = \frac{1}{\epsilon + d\sigma_j^2} \quad (39)$$

induces the same ordering as smaller σ_j^2 , smaller $\log \det T_{k,i}^{(j)}$, larger $\log \det (T_{k,i}^{(j)})^{-1}$, and larger $\text{tr}((T_{k,i}^{(j)})^{-1})$. The trace-based score therefore agrees with standard information criteria in the isotropic case; near-isotropic differences enter through anisotropy.

C.2. Algebraic Branch Structure of the Implemented GA-LMB Merger

The implemented merger uses separate spatial and existence weights because the GA-LMB update already exposes two different algebraic blocks. The spatial block is the Gaussian canonical update in (11), while the existence block is the Bernoulli geometric update in (13).

Proposition C2. For a fixed Bernoulli component i and conditional on w_x , the Gaussian density produced by (11) is the minimizer of

$$\min_{q \in \mathcal{G}} \sum_j w_{x,j} D_{\text{KL}}(q \parallel \mathcal{N}(v_{i,j}, T_{i,j})), \quad (40)$$

where $\mathcal{G}$ is the Gaussian family. Expanding the Gaussian KL in canonical parameters gives the precision and information-vector averages in (11), so the spatial update is the reverse-KL Gaussian barycenter.

Existence branch. Using (13), the same merger can be written in log-odds form.

Proposition C3. The fused existence probability satisfies

$$\log \frac{r_i}{1 - r_i} = \log \eta_i(w_x) + \sum_j w_{r,j} \log \frac{r_{i,j}}{1 - r_{i,j}}. \quad (41)$$

Thus, for fixed spatial weights, the existence branch is a weighted Bernoulli logit pool with an additive spatial-overlap offset.

C.3. From Bernoulli-RFS Decomposition to an Aligned-LMB Surrogate

Proposition C4. For Bernoulli RFS densities $\beta = (r, p)$ and $\beta_j = (r_j, p_j)$,

$$D_{\text{KL}}(\beta \parallel \beta_j) = d_{\text{Ber}}(r \parallel r_j) + r D_{\text{KL}}(p \parallel p_j), \quad (42)$$

where

$$d_{\text{Ber}}(r \parallel r_j) = (1 - r) \log \frac{1 - r}{1 - r_j} + r \log \frac{r}{r_j}. \quad (43)$$

The spatial mismatch is therefore naturally weighted by the existence level.

Proposition C5. For a fixed Bernoulli component i , consider the decoupled surrogate

$$\mathcal{J}_i(r, p; w_x, w_r) = \sum_j w_{r,j} d_{\text{Ber}}(r \parallel r_{i,j}) + r \sum_j w_{x,j} D_{\text{KL}}(p \parallel p_{i,j}). \quad (44)$$

Its minimizer uses the normalized geometric spatial product corresponding to (11), and its existence logit is the identity in (41). The spatial part follows by normalizing the weighted geometric product. Substituting this optimum gives the $-\log \eta_i(w_x)$ spatial contribution, and differentiating the Bernoulli part gives the logit identity. With aligned labels and product LMB form, the same argument lifts labelwise because the reverse-KL divergence is additive across aligned Bernoulli factors.

Scope boundary. This surrogate supports a compatibility statement: under the merger's label-alignment and projection assumptions, the implemented update matches a per-Bernoulli aligned-LMB objective.

C.4. Branch-Decoupled Weight Allocation and Structure-Aware Regularization

Let $\tilde{s}_{x,j}$ and $\tilde{s}_{r,j}$ denote the pre-normalization branch scores obtained from the dedicated scores and geometric interpolation in (21)–(23). This is convex interpolation in log-utility space. Define $\hat{u}_{x,j} = \log \tilde{s}_{x,j}$ and $\hat{u}_{r,j} = \log \tilde{s}_{r,j}$ before graph and FID-FIA refinement.

The FID-FIA refinement modifies only the existence branch. Let $q_{\text{fid},j}$ denote the bounded cue defined by (30)–(31). Using the existence-only modulation in (32), positive $q_{\text{fid},j}$ contributes $\gamma_f \log q_{\text{fid},j}$ to the existence utility and leaves the spatial utility unchanged. The main zero-floor setting is the boundary case in which a sensor with zero FID-FIA accumulation can receive zero instantaneous existence weight.

The structure-aware refinement then multiplies the spatial score by $\xi_{x,j}^{\gamma_x}$ and the existence score by $\xi_{r,j}^{\gamma_r}$, where $\xi_{x,j}$ and $\xi_{r,j}$ are positive branch-specific graph priors.

Proposition C6. Let p_x and p_r be normalized priors satisfying

$$p_{x,j} \propto \xi_{x,j}^{\gamma_x/\tau_x}, \quad (45)$$

$$p_{r,j} \propto \xi_{r,j}^{\gamma_r/\tau_r}. \quad (46)$$

Then the branch-wise allocations are equivalently written as

$$w_x^* = \arg \max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j \hat{u}_{x,j} - \tau_x D_{\text{KL}}(w \parallel p_x) \right\}, \quad (47)$$

$$w_r^* = \arg \max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j \hat{u}_{r,j} - \tau_r D_{\text{KL}}(w \parallel p_r) \right\}. \quad (48)$$

Thus the weak structure-aware term is a KL regularization toward a graph-induced prior on the simplex.

Corollary C6a. With the FID-FIA existence refinement, the same program remains valid after replacing $\hat{u}_{r,j}$ by $\hat{u}_{r,j}^{\text{fid}} = \hat{u}_{r,j} + \gamma_f \log q_{\text{fid},j}$. Equivalently,

$$w_{r,j}^* \propto \tilde{s}_{r,j} \xi_{r,j}^{\gamma_r} q_{\text{fid},j}^{\gamma_f}, \quad w_{x,j}^* \propto \tilde{s}_{x,j} \xi_{x,j}^{\gamma_x}. \quad (49)$$

The FID-FIA term is therefore an existence-branch log-utility that leaves the spatial KLA barycenter unchanged.

Proposition C7. Define

$$U_x(w_x) = \sum_j w_{x,j} \hat{u}_{x,j} + \tau_x H(w_x), \quad (50)$$

$$U_r(w_r) = \sum_j w_{r,j} \hat{u}_{r,j} + \tau_r H(w_r), \quad (51)$$

with the FID-FIA substitution in U_r when Cardinality-critical mode is active. Under the aligned-label and projection assumptions above, the branch-decoupled adaptive weights solve

$$\max_{w_x, w_r \in \Delta_{k,s}} \{U_x(w_x) + U_r(w_r)\}. \quad (52)$$

Together with Proposition C5, this gives a block-relaxed interpretation: the state-update block reproduces the aligned Bernoulli/LMB surrogate, and the weight-update block reproduces the branch-wise adaptive allocations.

Scalar-FID-FIA as a constrained special case. A single FID-FIA scalar weight for the whole fused posterior imposes $w_x = w_r = w_{\text{fid}}$ and lets the Fisher-separability utility influence both the Gaussian barycenter and the Bernoulli logit pool. The proposed refinement relaxes that constraint: w_x remains governed by the communication-aware spatial utility, while w_r receives the FID-FIA term. This explains why a scalar FID-FIA baseline can improve cardinality while perturbing localization; the branch-decoupled version reserves the information-geometric cue for existence/cardinality fusion.

C.5. Existence Confidence and Temporal Stabilization

The existence-confidence term uses $c(r) = |2r - 1|$, which is bounded in $[0, 1]$ and monotone in the magnitude of the Bernoulli natural parameter:

$$\left| \log \frac{r}{1-r} \right| = \log \frac{1+c(r)}{1-c(r)} = 2 \operatorname{artanh}(c(r)). \quad (53)$$

Thus existence confidence is a bounded proxy for the Bernoulli decisiveness appearing in (41).

The temporal stabilization step computes

$$w_k^{\text{ema}} = \alpha_{\text{EMA}} w_{k-1} + (1 - \alpha_{\text{EMA}}) \hat{w}_k. \quad (54)$$

Proposition C8. For any $\alpha_{\text{EMA}} \in [0, 1]$, w_k^{ema} is the minimizer of

$$\min_w \{ \alpha_{\text{EMA}} \|w - w_{k-1}\|_2^2 + (1 - \alpha_{\text{EMA}}) \|w - \hat{w}_k\|_2^2 \}. \quad (55)$$

The EMA step is therefore a quadratic compromise between the previous weights and the instantaneous weights. The subsequent minimum-weight enforcement is a pragmatic feasibility projection outside this quadratic interpretation.

CRedit authorship contribution statement

Hao Lang: Conceptualization, Supervision, Project administration, Writing - review and editing. **Jinhao Chen:** Software, Validation, Formal analysis, Writing - original draft. **Tianyu Wo:** Supervision, Writing - review and editing.

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Data availability

No real-world datasets were used in this study. The simulation data supporting the findings are synthetic and can be regenerated from the MATLAB/Octave scripts and deterministic seeds used in the experiments. The code and generated result logs will be made available by the corresponding author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT by OpenAI to improve the clarity, grammar, and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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